

baicalensis and the bark of *Acacia catechu*. The primary components of Flavocoxid are baicalin and catechin, concentrated and standardized to greater than 90% purity.²³⁰ It has been shown to inhibit activities of cyclooxygenase 1-, 2-, and 5-lipoxygenase²³¹⁻²³³ and was demonstrated to be safe²³⁰; however, its safety has not been evaluated when administered with other cyclooxygenase inhibitors such as NSAIDs or glucocorticoids. Caution should be exercised when used with concurrent cyclooxygenase inhibitors. It appears to be beneficial in humans with osteoarthritis.^{231,234}

Collagen Hydrolysates

Collagen hydrolysate is obtained by enzymatic hydrolysis of collagenous tissues from mammals, which is similar in amino acid composition to type II collagen, and provides high levels of glycine and proline that are involved in regeneration and stability of cartilage.^{82,235} Animal models showed benefits with treatment of induced osteoarthritis.²³⁶ In humans, it has been shown to improve joint pain and mobility and reduce analgesic medications in healthy active individuals with joint pain but without osteoarthritis, and in humans with mild to severe osteoarthritis.²³⁷⁻²⁴⁰

Creatine

Creatine is used in muscle for production of adenosine triphosphate (ATP), which provides energy for muscle contraction. Creatine is sometimes used by body builders and people who exercise with the intent to increase muscle mass and muscular energy. However, some recommend its use with arthritis, especially rheumatoid arthritis in humans. In rheumatoid arthritis skeletal muscle weakness often accompanies the arthropathy. In one study creatine supplementation to humans with rheumatoid arthritis increased serum and skeletal muscle creatine content, but failed to increase skeletal muscle creatine phosphate concentration or strength.²⁴¹ In another study of humans undergoing total knee replacement, creatine supplementation did not improve body composition or skeletal muscle strength.²⁴² Creatine supplementation has been shown to improve physical function, lower limb lean mass, and quality of life in postmenopausal women with knee osteoarthritis undergoing strengthening exercises.²⁴³

Devil's Claw

Harpagophytum procumbens (devil's claw) has been used as an analgesic, a remedy for fever and allergies.²⁴⁴ The major chemical constituents of *Harpagophytum* are iridoid glycosides, phytosterols, aromatic acids, and flavonoids. Glycosides found in the tubers of the plant appear to be the most therapeutically important constituents. Whole-plant extracts appear to have a better therapeutic effect than those prepared from isolated parts.²⁴⁴ A review of clinical trials using *H. procumbens* preparations for treatment of joint and lower back pain found that using extracts

containing 50 to 60 mg harpagoside daily gave more reliable results and were more effective at alleviating pain and improving mobility than extracts with lower amounts.^{245,246} An in vitro study using two rat cell lines found that the glycosides inhibited lipopolysaccharide-induced inducible nitric oxide and cyclooxygenase-2 expression via nuclear factor κ B suppression, thereby inhibiting inflammation.²⁴⁷ Devil's claw is chondroprotective, possibly because of inhibition of inflammatory mediators, including cyclooxygenase-2, leukotrienes, nitric oxide, tumor necrosis factor- α , and interleukin-1 β , and by inhibiting matrix metalloproteinases and elastase.^{247,248} (Note that devil's claw may increase gastric acidity; therefore caution is recommended when using this compound with NSAIDs or corticosteroids.)

Du Huo Ji sheng Wan

Du Huo Ji sheng Wan is a Chinese herbal product containing *Angelica Pubescentis*, Mulberry mistletoe, *Campanulaceae*, *Hoelen*, *Eucommiae*, *Angelica*, *Rehmannia*, cinnamon, licorice (*Glycyrrhiza*), and ginger. In a double-blind, controlled study comparing it with diclofenac, pain and function scores improved in both groups to the same extent but occurred more slowly in the Du Huo Ji sheng Wan treatment arm.²⁴⁹

Ginger

Ginger (*Zingiber officinale*), rich in gingerols and shogaols, has been advocated for treatment of a variety of diseases, including osteoarthritis because it suppresses prostaglandin and nitric oxide production.^{250,251} In one crossover study in humans significant improvement in symptoms for both ginger and ibuprofen groups was found before crossover; however, at the study's end there was no difference between ginger and placebo.²⁵² In another study it was found to be less effective than turmeric.²⁵³

Green Tea

Green tea (*Camellia sinensis*) is a rich source of polyphenols known as catechins, including epigallocatechin 3-gallate (EGCG), which has shown promise in treatment of arthritic diseases.²⁵⁴ Epigallocatechin 3-gallate has been shown in vitro and in vivo in rodent models of arthritis to have chondroprotective activity through modulation of interleukin-1 and -6 and decreased matrix metalloproteinase and nitric oxide activities.²⁵⁵⁻²⁵⁹

Homeopathy

Homeopathy is a form of alternative and complementary medicine based on vitalist philosophy, which interprets diseases and sickness as caused by disturbances in vital force or life force. Homeopathic practitioners treat patients using highly diluted preparations that follow the "law of similars" in which "let like be cured by like." Homeopathic remedies are prepared by serial dilution with shaking by

forceful striking on an elastic body (called *succussion*). Each dilution followed by succussion is assumed to increase effectiveness (called *potentization*). In humans with osteoarthritis, homeopathy has not been shown effective, has been found to be less effective than analgesic administration, or beneficial effects were attributed to the consultation and not to the homeopathic remedy.^{132,260-265} One homeopathic remedy, Zeel[®], has been evaluated in dogs (see Nutritional Compounds Evaluated in Dogs).

Methylsulfonylmethane

Methylsulfonylmethane (MSM) is a sulfur-containing compound found in plant and animal tissues. Three clinical trials in humans have shown positive results when compared with placebo either alone or in combination with glucosamine²⁶⁶⁻²⁶⁸; however, metaanalyses have failed to demonstrate a clinically effective reduction in pain or improvement in function.^{269,270}

Phytodolor N

Phytodolor N is a fixed combination of extracts from aspen leaves and bark (*Populus tremula*), common ash bark (*Fraxinus excelsior*), and golden rod herb (*Solidago vigaurea*). It appears to have antiinflammatory, antiedematous, antioxidative, and analgesic properties.^{271,272} Three studies in humans reported results in favor of Phytodolor N when compared with placebo, although results are skeptical as the manufacturer undertook the studies or used a higher dose.^{273,274}

Pine Bark Extract (Pycnogenol[®])

Pycnogenol[®] is a special standardized extract containing polyphenols and procyanidins from the bark of the French maritime pine (*Pinus pinaster*).⁸² It exhibits antiinflammatory and antioxidative activity as well as inhibits matrix metalloproteinase and inducible nitric oxide production.²⁷⁵ Studies in humans with osteoarthritis show a positive effect on decreasing pain and increasing function.²⁷⁶⁻²⁷⁹

Quercetin

Quercetin is a plant-derived flavonoid that may exhibit antiinflammatory and antioxidant properties via inhibition of tumor necrosis factor- α -mediated interleukin-8 and monocyte chemoattractant protein-1 expression.²⁸⁰ In two studies, it was not effective or only minimally effective in vivo or in vitro^{281,282}; however, in one study in humans with osteoarthritis or rheumatoid arthritis, it improved synovial fluid quality and pain scores when administered in combination with glucosamine and chondroitin sulfate.²⁸³

Resveratrol

Resveratrol is a type of natural phenol found in high concentration in grape skin and red wine. It has antiinflammatory, antioxidant, and chondroprotective properties such as increasing proteoglycan synthesis and chondrocyte

proliferation, and by suppressing interleukin-1, reactive oxygen species, p53-induced apoptosis, leukotriene B₄, prostaglandin E₂, tumor necrosis factor α , and matrix metalloproteinases, which may be beneficial for osteoarthritis.²⁸⁴⁻²⁸⁸ It acts, in part, by activating Sirtuin 1, a histone deacetylase, and Bcl2.^{289,290} Only in vitro studies have been performed.

Rose Hip and Seed Powder

The rose hip is the fruit of the rose plant that typically is red-to-orange, but ranges from dark purple to black in some species. It has antiinflammatory and antioxidative effects and has shown benefit in humans with rheumatoid arthritis or chronic pain.²⁹¹⁻²⁹⁴

Rosemary

Rosemarinus officinalis is a common spice and household plant that is also used for medicinal purposes. Experimentally in rodents, it reduces nociceptive pain in various kinds of induced pain.²⁹⁵ It has been studied as rho iso-alpha acids from hops, rosemary, and oleanolic acid and decreases prostaglandin production through inhibition of cyclooxygenase, thereby decreasing pain.^{296,297}

SKI 306X

SKI 306X is a purified extract from a mixture of three oriental herbal medicines (*Clematis mandshurica*, *Trichosanthes kirilowii*, and *Prunella vulgaris*) that has been widely used for the treatment of inflammatory diseases including knee osteoarthritis in humans. In vitro studies show SKI 306X or *Clematis* inhibits cartilage degradation, chondrocyte apoptosis, and production of tumor necrosis factor α and interleukin-1 β .²⁹⁸⁻³⁰³ Extracts from *Clematis vitalba* exhibit antiinflammatory, antinociceptive, and antipyretic effects in induced osteoarthritis in mice.³⁰⁴ In a double-blind, controlled study of 96 patients with knee osteoarthritis, patients receiving SKI 306X showed lower levels of pain and better function when compared with placebo.³⁰⁵ In another study of 249 patients with knee osteoarthritis, SKI 306X was equally as effective as diclofenac.³⁰⁶

Willow Bark Extract (*Salix* spp.)

Extracts from the bark of *Salix* species have been used for treatment of pain. The analgesic effect is attributed to salicin and/or its metabolite, salicylic acid.³⁰⁷ It should be administered cautiously or not at all with NSAIDs and glucocorticoids because of these compounds. The mechanisms of action of willow bark extract include inhibition of cyclooxygenase-2,³⁰⁸ lipoxigenase,³⁰⁹ cytokine release,^{308,309} and hyaluronidase,³¹⁰ and it may have antioxidant activity.^{309,311,312} Evaluation in humans with osteoarthritis or rheumatoid arthritis has yielded differing results regarding effectiveness. A moderate analgesic effect was shown in one study,³¹³ whereas in two other studies no effect was

found.³¹⁴ (Note that because willow bark extract contains the parent compound of aspirin, concurrent use with other NSAIDs or corticosteroids is not recommended. Also, the half-life of this compound is relatively long.)

Nutrition and Osteoarthritis in Cats

Osteoarthritis occurs commonly in cats³¹⁵⁻³¹⁹; however, there is a paucity of information regarding nutrition or nutritional compounds in management. In a prospective, blinded, parallel group study evaluating a diet high in n-3 fatty acids and supplemented with green-lipped mussel extract and glucosamine/chondroitin fed to cats with osteoarthritis over 9 weeks, there was an improvement in mobility of cats fed the supplemented diet and a decline in cats fed the control diet.³²⁰ Although other nutritional compounds are used in cats with osteoarthritis, none has been evaluated in a controlled manner.^{86,317,321,322}

Summary

In summary, this chapter provides guidelines on nutritional requirements for dogs undergoing rehabilitation and how nutrition can influence the healing process. Numerous supplements are available and used in small animal rehabilitation, and their efficacy should be evaluated using an evidence-based approach to define their use in rehabilitation.

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